Exam Date & Time: 24-Mar-2025 (11:30 AM - 12:30 PM)

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
MIDTERM THEORY EXAMINATION (REGULAR) MARCH, 2025
SIXTH SEMESTER OF B.TECH (CSE)

**DESIGN OF LANGUAGE PROCESSOR [CSE313]**

**Marks: 30** **Duration: 60 mins.**

**Section -1 (MCQs)**

**Answer all the questions.**

Instructions:
1. Attempt all questions.
2. Assume that all given grammars are in Context-Free Grammar (CFG) unless it is not specified.

| 1 | | |
|---|---|---|

Consider the following lex specification rules:

%%

a?b { printf("Matched rule 1"); }

a?b { printf("Matched rule 2"); }

%%

If the input string is "a?b", which of the following statementsis correct?

(1)

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1) | Only rule 1 will match because the ? operator makes the preceding a optional, effectively matching "b" or "ab", which covers the input. | | 2) | Only rule 2 will match because the escaped ? in rule 2 matches a literal ? character, correctly matching "a?b". | | 3) | Both rules will match, causing a conflict that is resolved by lex’s longest match rule. | | 4) | Neither rule will match the input. | |

Course Outcome: "CO1" / Cognitive Level: "Remember"

| 2 | | |
|---|---|---|

A grammar with production rules { A → Ba | Cb, B → CA, C → c | ε }contains

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1) | Left Recursion | | 2) | Left Factor | | 3) | Both left factor and left recursion | | 4) | None of the above | |

(1)

Course Outcome: "CO2" / Cognitive Level: "Understand"

| 3 | | |
|---|---|---|

In SLR parsing for the grammar

$S \rightarrow Z \mid SabS$

$Z \rightarrow bZ \mid \varepsilon$

In state 0, for inputs 'a' and 'b':

(2)

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1) | Both will have shift-reduce conflict | | 2) | Only ‘a’ will have shift-reduce conflict | | 3) | Only ‘b’ will have shift-reduce conflict | | 4) | None of the above | |

Course Outcome: "CO3" / Cognitive Level: "Understand"

| 4 | | |
|---|---|---|

Match the following

| | |
|---|---|
| (a) Linear representations | (i) Syntax trees and DAGs |
| (b)Graphical representations | (ii) Quadruples, Triples, Indirect Triples |
| | (iii) Postfix notation |

(1)

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1) | a-i , b-ii | | 2) | a-iii , b-i | | 3) | a-ii , b-iii | | 4) | a-ii , b-ii | |

Course Outcome: "CO4" / Cognitive Level: "Remember"

## Section - 2 (Descriptive)

**Answer 2 out of 3 questions.**

Instructions:
1. Attempt any 2 questions out of 3.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Assume that all given grammars are in Context-Free Grammar (CFG) unless it is not specified.

| 1 | | |
|---|---|---|

Consider the expression a = b * c + 50 - d and describe how acompiler processes this expression. Demonstrating the input andoutput of each compilation phase-including lexical analysis, syntaxanalysis, semantic analysis, intermediate code generation and codeoptimization (Consider the Precedence as * > + > - ). (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

| 2 | | |
|---|---|---|

Consider the following grammar for languages built from terminals aand b, and prove whether it is LL(1) grammar or not by constructingLL(1) Parsing Table.

S -> C a C b A

A -> B

A -> S

B -> ϵ

B -> b

C -> ϵ

(5)

Course Outcome: "CO3" / Cognitive Level: "Application"

| 3 | | |
|---|---|---|

Construct the SLR (1) parsing table for the following grammar andshow a trace of a parsing of w = iii * i + * (5)

S -> i | SS+ | SS*

Course Outcome: "CO3" / Cognitive Level: "Application"

**Section - 3 (Descriptive)**

**Answer 3 out of 4 questions.**

Instructions:
1. Attempt any 3 questions out of 4.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Assume that all given grammars are in Context-Free Grammar (CFG) unless it is not specified.

| 1 | | |
|---|---|---|

Consider the following Grammars: Remove left recursion from thegrammar in part (a), Remove left factoring from the grammar in part(b). Why is it necessary to remove left recursion and perform leftfactoring when preparing a grammar?

(a) Remove Left Recursion
A -> ABd | Aa | a
B -> Be | b

(b) Remove Left Factor
A -> aAB | aA | a
B -> bB | b

(5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

| 2 | | |
|---|---|---|

Construct the operator precedence parsing table for the belowgrammar. Using the parsing table, show step-by-step parsing ofinput string " ((a), ^) ".

S -> a | ^ | (T)
T -> T,S | S

(5)

Course Outcome: "CO3" / Cognitive Level: "Understand"

| 3 | | |
|---|---|---|

Write a Syntax-Directed Definition (SDD) for the below grammar thatannotates each term (T) and expression (E) with its type (integeror float). Assume an input string as 3 + 5 + 2.5. Construct theannotated parse tree that demonstrates how the data typeinformation is computed and propagated throughout the derivation.

E -> E + T | T

T -> num **.** num | num

(5)

Course Outcome: "CO4" / Cognitive Level: "Application"

| 4 | | |
|---|---|---|

Translate the arithmetic expression (b+c) *a + −(b+c) into thefollowing representation:

a) A syntax tree.

b) Quadruples.

c) Triples.

d) Indirect triples

e) DAG

(5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

-----End-----

| Event Name | Subject |
|---|---|
| CHARUSAT Midterm Exam 2024-25 Even | CSE313:DESIGN OF LANGUAGE PROCESSOR |

| # | Course Outcome | Description | Marks |
|---|---|---|---|
| 1 | CO1 | Understand the fundamentals of the compiler identify the relationships among different phases of the compiler and use the knowledge of the Lex tool | 6 |
| 2 | CO2 | Describe theRole of Parser and the various error recovery strategies | 6 |
| 3 | CO3 | Develop the parsers and experiment with the knowledge of different parser designs. | 17 |
| 4 | CO4 | Develop a semantic analysis scheme to generate intermediate code | 11 |
| | | **Total** | **40** |

* NA - CO Not Selected

| Event Name | Subject |
|---|---|
| CHARUSAT Midterm Exam 2024-25 Even | CSE313:DESIGN OF LANGUAGE PROCESSOR |

| # | Level | No of Questions | Marks (%) |
|---|---|---|---|
| 1 | Remember | 2 | 2(5%) |
| 2 | Understand | 6 | 23(57.5%) |
| 3 | Application | 3 | 15(37.5%) |
| 4 | Analysis | 0 | 0(0%) |
| 5 | Create | 0 | 0(0%) |
| 6 | Evaluate | 0 | 0(0%) |
| | Total | 11 | 40 (100%) |